Toward an Emergent Origin of Fundamental Constants in General Emergence Mechanics (GEM)

c, G, $\hbar$, and k_B as Persistence-Mechanical Targets

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Abstract

We explore the possibility that several fundamental constants of physics — including the speed of light c, Newton's gravitational constant G, Planck's constant $\hbar$, and Boltzmann's constant k_B — may be interpretable as emergent features of the coherence shell in General Emergence Mechanics (GEM). Rather than treating these constants as wholly independent primitive inputs, we examine how they might arise as effective quantities associated with shell propagation, depth-gradient coupling, stochastic fluctuation scale, and fluctuation-to-thermal conversion. The paper does not claim completed derivations for all four constants. Instead, it organises them into a common persistence-based framework and clarifies the current derivational status of each. In this picture, a small number of shell parameters — including the vacuum scale Λ, orbifold bias ϵ, preferred phase $\phi_0 = 255°$, and coupling strength g — may ultimately underlie quantities that standard theory treats as independent. This framework suggests a possible route toward a more economical description of nature grounded in persistence mechanics.

Keywords: General Emergence Mechanics, GEM, emergent constants, speed of light, gravitational constant, Planck constant, Boltzmann constant, coherence shell, persistence mechanics, vacuum structure

Note. This paper is a synthesis and research-programme note rather than a claim of completed derivation. Its aim is to organise several speculative but structured routes by which fundamental constants might emerge from the coherence shell in GEM.

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1 Introduction

Modern fundamental physics contains a remarkable set of constants whose values are measured but not, in the usual formulation, deeply explained. These include the speed of light c, Newton's gravitational constant G, Planck's constant $\hbar$, and Boltzmann's constant k_B. In standard formulations, such quantities are indispensable, but they appear largely as inputs rather than consequences.

General Emergence Mechanics (GEM) suggests a different possibility. In GEM, stable physical structures arise through persistence within a dynamical coherence shell anchored at a preferred vacuum phase. If this shell has its own characteristic scales, response coefficients, fluctuation amplitudes, and thermalisation structure, then some of the constants conventionally treated as fundamental may instead be emergent features of that shell.

The purpose of the present note is to organise that possibility into a single framework. The claim is not that all major constants have already been derived. Rather, the claim is that GEM provides a natural language in which such derivations become meaningful targets. In particular:

- c may be associated with the propagation speed of coherent shell disturbances,
- G may be associated with the strength of depth-gradient coupling,
- $\hbar$ may be associated with the stochastic action scale of shell fluctuations,
- k_B may be associated with the conversion between shell fluctuations and macroscopic thermal behaviour.

If even part of this programme succeeds, then the number of apparently independent constants in physics may be reduced. Constants would become different projections of the same persistence mechanics rather than isolated primitive facts.

2 The Coherence Shell as a Constant-Generating Structure

In GEM, the vacuum is not treated as an inert backdrop. It is a structured coherence shell with a preferred minimum at

$$\phi_0 = 255^\circ = \frac{17\pi}{12}.$$

The shell is characterised, at effective level, by a small set of parameters such as:

- the ultraviolet or vacuum scale Λ,
- the orbifold bias parameter ϵ,
- the preferred phase ϕ_0,
- one or more coupling strengths g,
- effective dissipation or fluctuation parameters associated with unresolved shell dynamics.

The central hypothesis of this paper is that constants may arise not as unrelated primitives but as derived or effective quantities built from this smaller shell parameter set.

At a broad conceptual level, the logic is as follows:

- shell propagation properties may generate c,
- shell-depth coupling may generate G,

- shell fluctuation amplitudes may generate $\hbar$,
- shell fluctuation-to-temperature conversion may generate k_B.

This is a research programme, not a finished reduction. But it provides a unified target architecture.

3 Current Derivational Status

Not all constants stand on the same footing in the current state of GEM development. Table 1 summarises their present status.

Table 1: Current derivational status of candidate emergent constants in GEM.

Constant	GEM interpretation	Current status
c	Shell propagation speed $\sqrt{\lambda/\chi}$	Structurally developed, not yet numerically fixed
G	Depth-gradient coupling strength	Partial effective interpretation
$\hbar$	Stochastic shell action scale	Speculative but mathematically suggestive
k_B	Fluctuation-to-thermal conversion scale	Speculative but naturally motivated

This uneven maturity is important. The purpose of the paper is not to hide it, but to clarify it.

4 The Speed of Light as a Shell Propagation Speed

The most developed constant in the present framework is the speed of light. Near the preferred vacuum minimum, the effective shell Lagrangian may be written in the form

$$\mathcal{L} = \frac{\chi}{2}(\partial_t \phi)^2 - \frac{\lambda}{2}(\nabla\phi)^2 - V(\phi),$$

where χ is the temporal inertia coefficient and λ is the spatial gradient stiffness.

Linearising around ϕ_0 gives

$$\chi\, \partial_t^2 \delta\phi - \lambda\, \nabla^2 \delta\phi + \kappa_{\text{eff}} \delta\phi = 0, \qquad \kappa_{\text{eff}} = V''(\phi_0).$$

For Fourier modes

$$\delta\phi \propto e^{i(\mathbf{k}\cdot\mathbf{x} - \omega t)},$$

the dispersion relation is

$$\chi\omega^2 = \lambda k^2 + \kappa_{\text{eff}}.$$

The local curvature term κ_{eff} sets the restoring or mass scale, but the characteristic propagation speed is controlled by the ratio of spatial gradient stiffness to temporal inertia. In the high-k or weak-mass limit, the group velocity approaches

$$v_{\text{group}} \to \sqrt{\frac{\lambda}{\chi}}.$$

This suggests the identification

$$\boxed{c_{\text{shell}}^2 = \frac{\lambda}{\chi}}.$$

The important point is not yet the exact numerical match, but the structural one. GEM naturally contains a shell propagation speed with exactly the right formal role to be compared with the observed speed of light. This makes c a genuine derivational target within the framework rather than a separate unexplained primitive.

5 Newton's Gravitational Constant as a Depth-Coupling Strength

The next constant with partial structure in GEM is Newton's gravitational constant G. In the gravitational branch of the framework, gravity is associated with spatial gradients in persistence depth:

$$\mathbf{F}_{\text{eff}} \propto -\nabla D.$$

This suggests that the effective strength of gravitational coupling is controlled by how efficiently matter sources local depth deformation, and how strongly depth gradients feed back into the motion of persistent configurations.

At exploratory level, one may therefore interpret G as measuring the efficiency of depth-gradient coupling. Schematically,

$$G \sim \frac{1}{D_0} \times (\text{dimensionful shell factors}),$$

where D_0 is a reference depth scale and the omitted factors arise from the properly normalised shell action, matter coupling, and unit restoration.

This does not yet amount to a derivation of the observed numerical value of G. However, it provides a clear physical interpretation:

> G may be the effective strength with which matter deepens the shell and with which those depth gradients influence persistent motion.

In that sense, G would no longer be a primitive coupling, but a measure of vacuum persistence responsiveness.

6 Planck's Constant as a Stochastic Action Scale

A more speculative but mathematically suggestive possibility concerns Planck's constant $\hbar$. In the stochastic depth formulation of GEM, the shell is described not only by mean structure but also by fluctuations:

$$\frac{\partial D}{\partial t} = -\gamma \frac{\delta F[D]}{\delta D} + \xi(\mathbf{x}, t),$$

with noise correlations of the general form

$$\langle \xi(\mathbf{x}, t)\xi(\mathbf{x}', t')\rangle \propto \delta(\mathbf{x} - \mathbf{x}')\delta(t - t').$$

This raises the possibility that a fundamental action scale analogous to $\hbar$ may emerge from the amplitude and coherence scale of shell fluctuations. Schematically, one may imagine a relation of the form

$$\hbar_{\text{eff}} \sim (\text{depth fluctuation amplitude}) \times (\text{coherence length}) \times (\text{coherence momentum scale}),$$

or more compactly as an emergent fluctuation-action scale of the shell.

At present this is not a derivation. It is a structured conjecture. Its attraction lies in the fact that quantum behaviour in GEM already points toward a fluctuation sector, and $\hbar$ is precisely the constant that sets the scale of quantum action. If that action scale could be tied to stochastic shell structure, then quantum mechanics itself would become less primitive and more emergent.

The immediate value of this idea is therefore conceptual: it identifies a natural place in the GEM framework where an effective $\hbar$-like quantity might arise.

7 Boltzmann's Constant as a Fluctuation-to-Thermal Scale

A particularly natural constant in the stochastic shell picture is Boltzmann's constant k_B. In standard statistical physics, k_B sets the conversion scale between temperature and microscopic energy. In GEM, this suggests an immediate interpretation: k_B may be the effective conversion factor between stochastic depth fluctuations and macroscopic thermal behaviour.

The stochastic depth field evolves schematically as

$$\frac{\partial D}{\partial t} = -\gamma \frac{\delta F[D]}{\delta D} + \xi(\mathbf{x}, t),$$

with noise satisfying a fluctuation-dissipation-style relation of the form

$$\langle \xi(\mathbf{x}, t)\xi(\mathbf{x}', t')\rangle = 2\gamma k_B T\, \delta(\mathbf{x} - \mathbf{x}')\delta(t - t').$$

In the GEM picture, such a fluctuation-dissipation relation is a natural candidate consequence of the shell's stochastic dynamics rather than a wholly independent input. This suggests that k_B may be interpretable as the scale relating:

- the variance of persistence-depth fluctuations,
- the dissipation scale of the shell,
- and the effective thermal description of macroscopic behaviour.

At schematic level, one expects an effective relation of the form

$$k_B^{\text{eff}} \sim \frac{\langle (\delta D)^2 \rangle}{\Delta T},$$

where $\langle (\delta D)^2 \rangle$ is the variance of persistence-depth fluctuations and ΔT is the corresponding thermal scale.

Likewise, matching the stochastic shell description to a fluctuation-dissipation form suggests an effective thermal conversion scale of the schematic form

$$k_B^{\text{eff}} \sim \frac{\gamma_{\text{eff}} \langle \xi^2 \rangle}{2T},$$

up to normalisation conventions and the precise definition of the noise correlator.

This interpretation is attractive because it makes k_B conceptually parallel to the other constants:

- c measures propagation,
- G measures depth coupling,
- $\hbar$ measures fluctuation action scale,
- k_B measures fluctuation-to-temperature conversion.

One further suggestive consequence is that in regions of altered shell depth, the local fluctuation structure may also change. This raises the possibility of GEM-native routes toward Tolman-like thermal behaviour in gravitationally deep regions, though such links remain speculative at present.

8 Toward a Reduction of Free Parameters

The larger significance of this programme lies in its potential to reduce the number of primitive inputs in physics. If c, G, $\hbar$, and perhaps eventually k_B are all emergent from shell structure, then a substantial part of what is currently treated as arbitrary may instead be derivable from a much smaller set of parameters:

- vacuum scale Λ,
- orbifold bias ϵ,
- preferred phase ϕ_0,
- coupling strength g,
- and perhaps one or two fluctuation or dissipation parameters.

This would not yet remove all free parameters from physics, but it would mark a major conceptual shift. Constants would become shell observables rather than separate fundamental givens.

9 Falsifiability and Open Problems

For this programme to be more than a suggestive synthesis, it must eventually become empirically and mathematically vulnerable. The main open problems are:

- extracting λ and χ explicitly from the properly normalised MT/GCF shell action and determining whether c_{shell} agrees quantitatively with the observed c,
- deriving the depth-coupling strength rigorously enough to compare with the observed value of G,
- determining whether a fluctuation-induced action scale can genuinely reproduce the role of $\hbar$,
- clarifying whether stochastic shell fluctuations produce a natural thermal conversion scale matching k_B,
- and establishing whether the reduced shell parameter set is actually sufficient to recover more than one constant at once.

A particularly important empirical direction would be to look for correlations or constraints suggesting that constants ordinarily treated as independent may share a common origin in shell structure. Even partial success here would be significant.

10 Conclusion

This paper has organised a unified GEM-based research programme for the possible emergence of several fundamental constants. The strongest current case is the speed of light, which appears naturally as a shell propagation speed

$$c_{\text{shell}}^2 = \frac{\lambda}{\chi}.$$

Newton's constant G has a partial effective interpretation as a depth-coupling strength. Planck's constant $\hbar$ has a plausible but still speculative route through stochastic shell fluctuations. Boltzmann's constant k_B has a natural candidate interpretation as the conversion scale between shell fluctuations and thermal behaviour.

The paper does not claim that all of these constants have already been derived. What it claims instead is that GEM provides a common persistence-mechanical language in which their emergence can be meaningfully investigated. If that programme succeeds even partially, then the constants of nature may turn out to be less fundamental and more emergent than previously supposed.

References

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